Friday

1. Start 10.3 Polar Coordinates
2. Polar Calculus (tangent lines)

Monday

1. Start 10.4 Polar Area

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⑥ Find length of one cycle of $\quad dS = \sqrt{\left(\frac{dy}{dt}\right)^2 + \left(\frac{dx}{dt}\right)^2}$

$x = 3(t - \sin t) \quad y = 3(1 - \cos t)$

$$dS = \sqrt{(3(1-\cos t))^2 + (3(\sin t))^2}\, dt$$

$$= \sqrt{9(1 - 2\cos t + \cos^2 t) + 9\sin^2 t}\, dt$$

$$= \sqrt{9 - 18\cos t + \underbrace{9\cos^2 t + 9\sin^2 t}_{9}}\, dt$$

$$= \sqrt{18 - 18\cos t}\, dt = \sqrt{9(2 - 2\cos t)}\, dt = 3\sqrt{2 - 2\cos t}\, dt$$

$$\int 3\sqrt{2 - 2\cos t}\, dt$$

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4. $x = t^2 - 16t - 6 \qquad y = t^2 + 16t - 6$

$\frac{dx}{dt} = 2t - 16 = 0 \qquad \frac{dy}{dt} = 2t + 16 = 0$

$t = 8 \qquad t = -8$

horiz tangent

$x = 8^2 - 16 \cdot 8 - 6 = -70 \qquad x = 186 \qquad (186, -70)$

$y = -70$

$y = 8^2 + 16 \cdot 8 - 6 = 186$

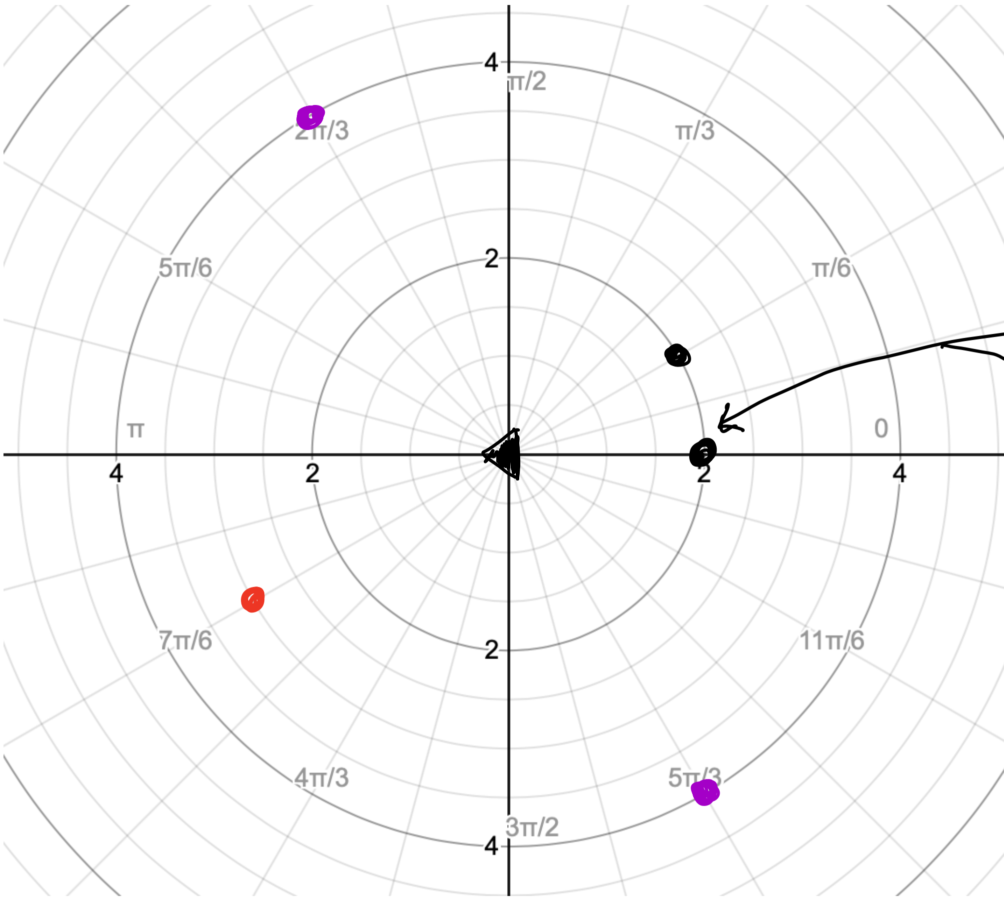
Vert tan at $(-70, 186)$

# 10.3 Polar Coordinates

## 1 Definition of Polar Coordinates

$\left(3, \frac{7\pi}{6}\right)$

We can specify a point in the plane by telling its distance $r$ from the origin, and the angle $\theta$ that the point lies on. The ordered pair $(r, \theta)$ is the set of **polar coordinates** of the point.




$\pi$

$\left(2, \frac{\pi}{6}\right)$

$\left(4, \frac{2\pi}{3}\right)$

$(-2, \pi)$

$(2, 0)$

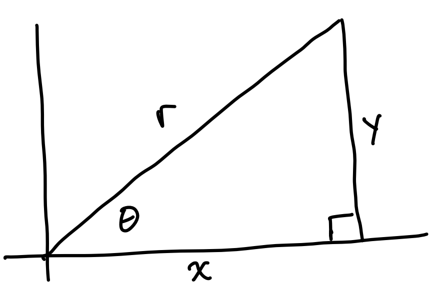
$\left(-4, \frac{2\pi}{3}\right)$

In Rectangular (aka Cartesian) coordinates, every point has exactly one set of coordinates.
In Polar coordinates, each point has infinitely many sets of coordinates.

**Rules:**

- The points $(r, \theta)$ and $(r, \theta + 2\pi n)$ are equal for all integer values of $n$.
- The points $(r, \theta)$ and $(-r, \theta + \pi)$ are equal.
- If $r = 0$, then the point $(0, \theta)$ is the origin for any value of $\theta$.

## 2 Converting Points Between Polar and Rectangular




$\boxed{x^2 + y^2 = r^2 \qquad \tan\theta = \frac{y}{x}}$ Rect → Polar

$\boxed{x = r\cos\theta \qquad y = r\sin\theta}$ Polar → Rect

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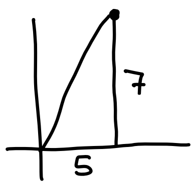
### 2.0.1 Example:

Convert $x = 5, y = 7$ to polar. Give answer in both degrees and radians.




$r^2 = 5^2 + 7^2 = 25 + 49 = 74$

$r = \sqrt{74}$

$\tan\theta = \frac{7}{5}$ $\quad \theta = 54.46^\circ$

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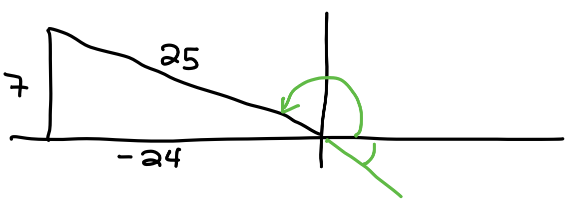
### 2.0.2 Example:

Convert $x = -24, y = 7$ to polar. Give answer in both degrees and radians.




$r = \sqrt{7^2 + (-24)^2} = 25$

$\tan\theta = \frac{7}{-24}$

$\theta_{Ref} = -16.26^\circ$

$+ 180$

$\theta = 163.74^\circ$

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### 2.0.3 Example:

Convert $\left(4, \frac{3}{4}\pi\right)$ to rectangular.

# 3 Converting Equations Between Cartesian and Polar

We still use

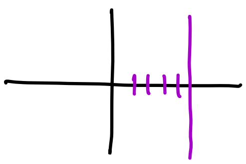
$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad x^2 + y^2 = r^2, \qquad \text{and} \qquad \tan\theta = \frac{y}{x}$$

### 3.0.1 Example:

Convert $x = 5$ to Polar.

$$5 = r\cos\theta$$

$$r = \frac{5}{\cos\theta}$$



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### 3.0.2 Example:

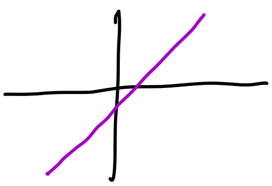
Convert $y = x - 1$ to Polar.

$$r\sin\theta = r\cos\theta - 1$$

$$r\sin\theta - r\cos\theta = -1$$

$$r(\sin\theta - \cos\theta) = -1$$

$$r = \frac{-1}{\sin\theta - \cos\theta}$$



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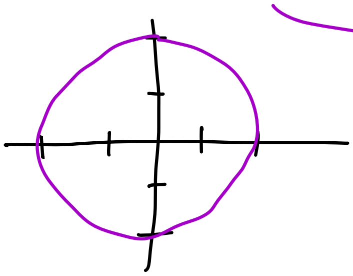
### 3.0.3 Example:

Convert $x^2 + y^2 = 4$ to Polar.

$$x^2 + y^2 = r^2$$

$$r^2 = 4$$

$$r = 2$$



# 4 Finding the slope of a tangent line

Given $r = f(\theta)$, we treat $\theta$ as a parameter:

$$x = r\cos\theta \qquad \text{and} \qquad y = r\sin\theta$$

Replacing $r$ with $f(\theta)$ in these gives us

$$x = f(\theta)\cos\theta \qquad \text{and} \qquad y = f(\theta)\sin\theta$$

From here we get

$$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}}$$

### 4.0.1 Example:

(a) Find the equation of the line tangent to $r = \underline{1 + \cos(\theta)}$ at $\theta = \frac{\pi}{2}$

(b) Where does the graph have horizontal and vertical tangents?

(a) $x = r\cos\theta$

$x = (1+\cos\theta)\cos\theta$

$x = \cos\theta + \cos^2\theta$

$\frac{dx}{d\theta} = -\sin\theta - 2\cos\theta\sin\theta$

$\left.\frac{dx}{d\theta}\right|_{\theta=\frac{\pi}{2}} = -\sin\frac{\pi}{2} - 2\cos\frac{\pi}{2}\sin\frac{\pi}{2}$

$-1 \quad - 2\cdot 0\cdot 1$

$= -1$

$y = r\sin\theta$

$y = (1+\cos\theta)\sin\theta$

$y = \sin\theta + \cos\theta\sin\theta$

$\frac{dy}{d\theta} = \cos\theta + \cos\theta\cos\theta - \sin\theta\sin\theta$

$\cos\theta + \cos^2\theta - \sin^2\theta$

$\left.\frac{dy}{d\theta}\right|_{\theta=\frac{\pi}{2}} = 0 + 0^2 - 1^2 = -1$

$\text{slope} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{-1}{-1} = 1$

At $t = \frac{\pi}{2}$, $x = \cos\frac{\pi}{2} + \cos^2\frac{\pi}{2} = 0$

$y = \sin\frac{\pi}{2} + \cos\frac{\pi}{2}\sin\frac{\pi}{2} = 1$

$1 \quad + 0$

$y - 1 = 1(x - 0)$

$\boxed{y = x + 1}$